

Importanza dell'Usabilità nel Processo di Design

Corso di Informatica – Laurea Magistrale

Prof.ssa Giuliana Vitiello

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Outline

- The changing face of design
- Usability as a major concern in design
- Design as part of UE
- User-centred design: from conceptual design to scenario-based design



The Changing Face of Design

- In recent years, the role of computers in society has changed radically.
- Technological evolution also brings with it new custom design activities
- And a new body of design knowledge is being built:
 - systems for ubiquitous computing
 - small communities of specific users
 - social participation
 - user-provided context (from Web 2.0 onwards)
 - software for service provision
 - collaborative environments...
 - ...and many others



The Changed Focus for Design

From

- defining system operations (e.g., functional specifications)

To

- describing how people will **use** a system to accomplish work tasks and other activities

Different Design Approaches

- professionally-dominated design
 - works best for people with the same interests and background knowledge
- user-centered design
 - analyze the needs of the users
 - understand the conceptual world of the users
- learner-centered design
 - draw attention to the changing needs of the user
 - combines HCI interaction principles with educational interaction support
- participatory design
 - involves users more deeply in the process as co-designers
 - focus on system development at design time

Overall...

... focus is on the

USER

and on

USER'S NEEDS





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Usability as a major concern in design

- ISO 9241-11 *Ergonomics of Human System Interaction - Guidance on Usability* (1998)

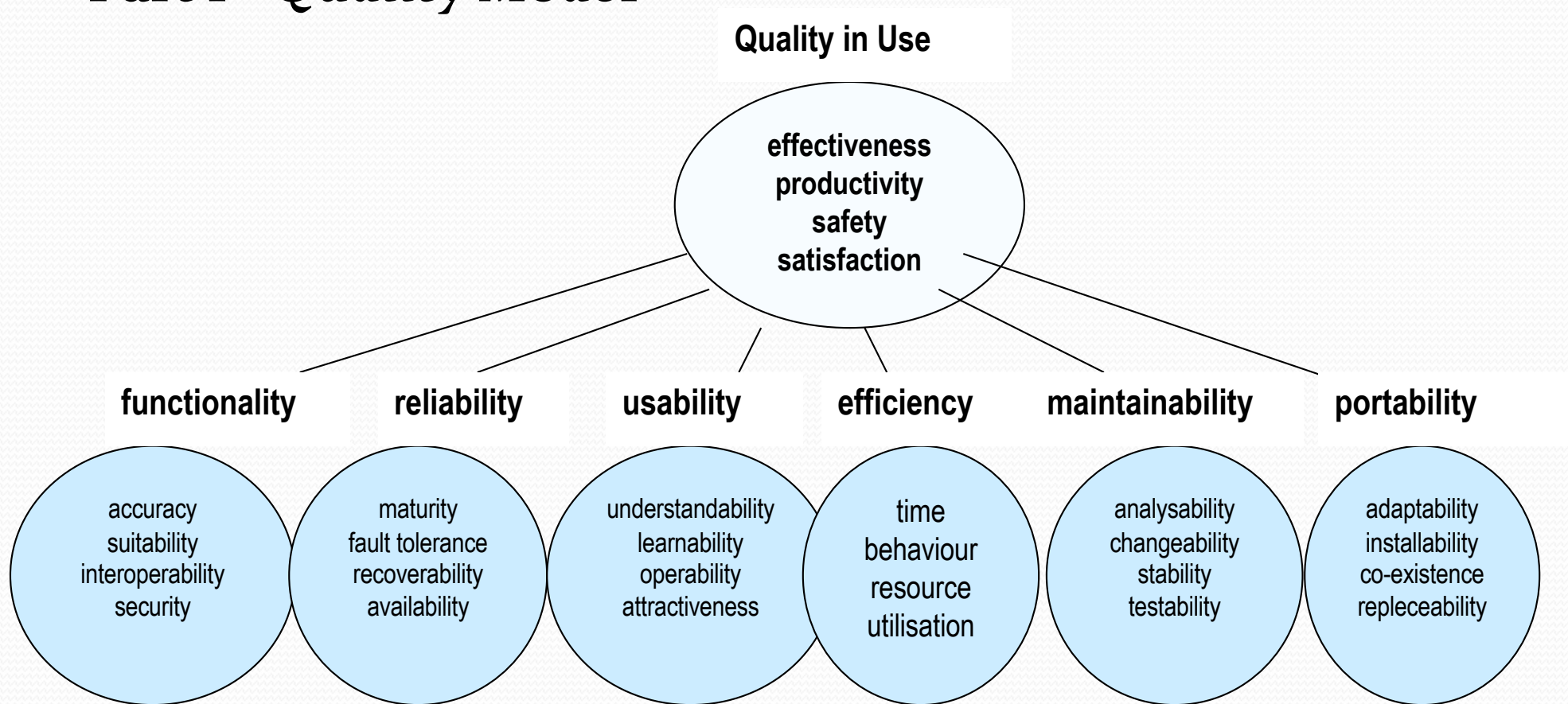
Usability is “*the extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use*”

- *Usability of a software product is one of the factors contributing to **quality in use**.*

Usability as a major concern in design (ctd)

- ISO/IEC-9126 (2001) standard on *Software Engineering — Product Quality*

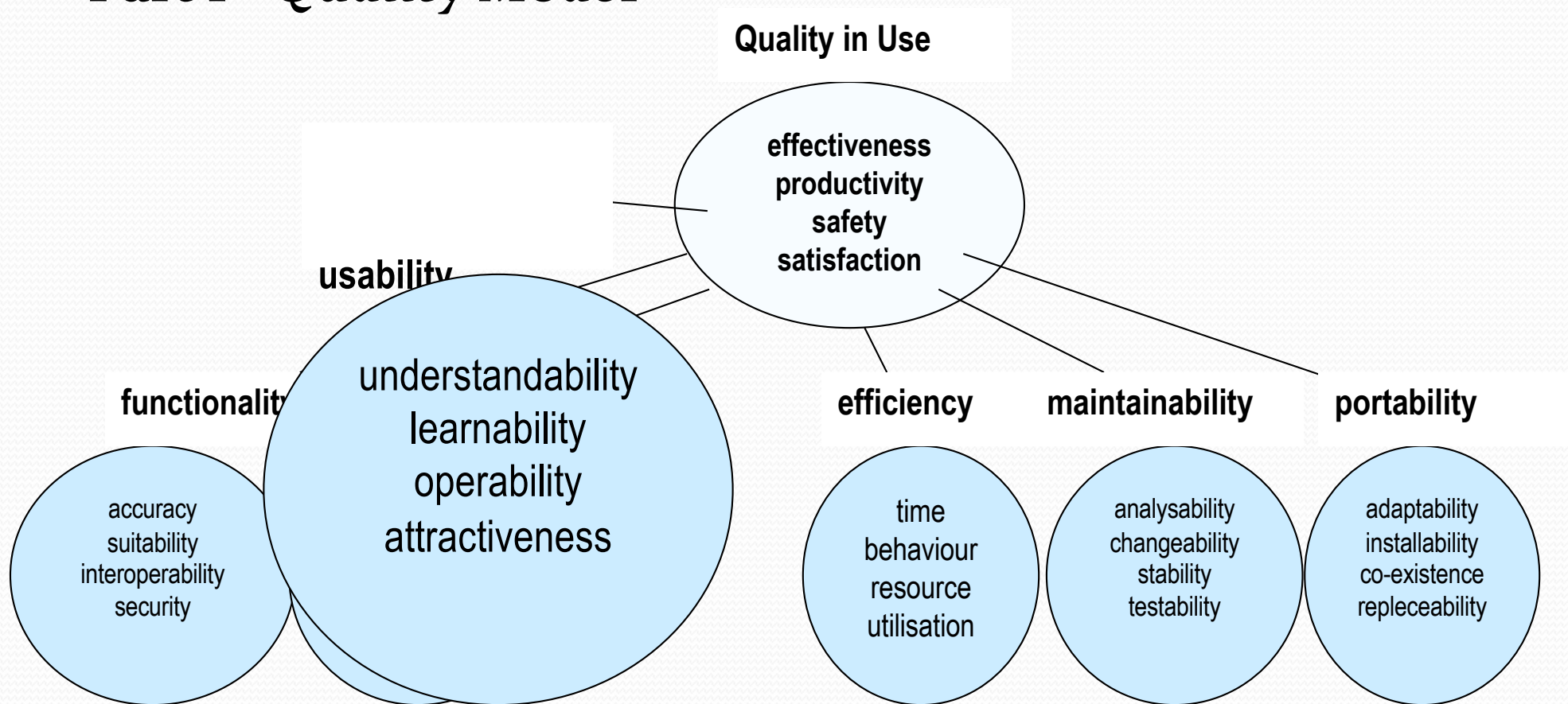
Part I - Quality Model



Usability as a major concern in design (ctd)

- ISO/IEC-9126 (2001) standard on *Software Engineering — Product Quality*

Part I - Quality Model



Usability Dialogue principles

- ISO 9241-110:2006 Ergonomics of human-system interaction -- Part 110: Dialogue principles.
- Lifecycle Phase
 - Create the user experience.
- A set of usability heuristics that applies to the interaction of people and information systems
- The standard refers to this interaction as a “dialogue” and describes seven “dialogue principles”.

The seven principles of ISO9241-110

1. suitability for the task (the dialogue should be suitable for the user's task and skill level);
2. self-descriptiveness (the dialogue should make it clear what the user should do next);
3. controllability (the user should be able to control the pace and sequence of the interaction);
4. conformity with user expectations (it should be consistent);
5. error tolerance (the dialogue should be forgiving);
6. suitability for individualisation (the dialogue should be able to be customised to suit the user);
7. suitability for learning (the dialogue should support learning).

ISO 9241-11	Shneiderman	Nielsen
efficiency	Speed of performance	Efficiency
	Time to learn	Learnability
effectiveness	Retention over time	Memorability
	Rate of errors by users	Errors/Safety
satisfaction	Subjective satisfaction	Satisfaction

Cognitive skills of users

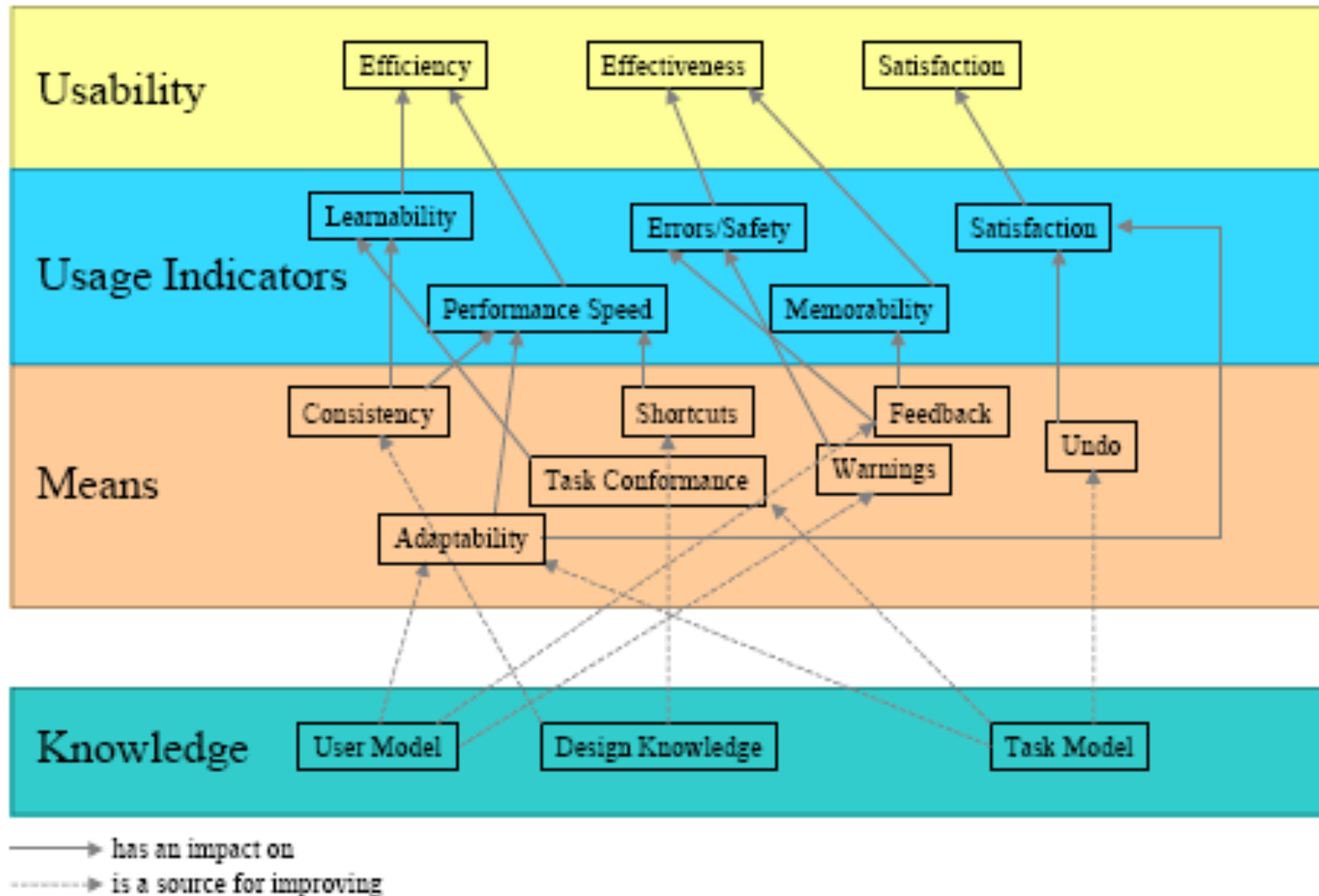
efficiency

effectiveness

Learnability	Flexibility	Robustness
Predictability	Dialog initiative	Observability
Synthesizability	Multi-Threading	Recoverability
Familiarity	Task Migratability	Responsiveness
Generalizability	Substitutivity	Task conformance
Consistency	Customizability	

Alan Dix

Layered Model of Usability(*)



Van Welie, M. and Van der Veer, G. (1999). "Ontologies and Methods in Interdisciplinary Design. Computer Science Education", Challenges for the new Millenium, pp. 143-158, Casa Cartii de Stiinta.

Interface design factors affecting *learnability*

- Predictability
 - User-centered – Perceived affordance
- Synthesizability
 - Visibility
- Familiarity
 - Metaphors
- Generalizability
 - Uniform model
- Consistency

Interface design factors affecting *flexibility*

Dialogue initiative

- freedom from system imposed constraints on input dialogue
- system vs. user pre-emptiveness

Multithreading

- ability of system to support user interaction for more than one task at a time
- concurrent vs. interleaving; multimodality

Task migratability

- passing responsibility for task execution between user and system

Substitutivity

- allowing equivalent values of input and output to be substituted for each other
- representation multiplicity; equal opportunity

Customizability

- modifiability of the user interface by user (adaptability) or system (adaptivity)

Dix's concrete point of view

Interface design principles affecting *robustness*

- **Observability** – user's capability to evaluate the internal state of the system from its perceivable representation.
 - *browsability; defaults; reachability; persistence; operation visibility*
- **Recoverability** – ability of user to take corrective action once an error has been recognized
 - *reachability; forward/backward recovery; commensurate effort*
- **Responsiveness** – how the user perceives the rate of communication with the system
- **Task Conformance** - degree to which system services support all of the user's tasks
 - *task completeness; task adequacy*



Outline

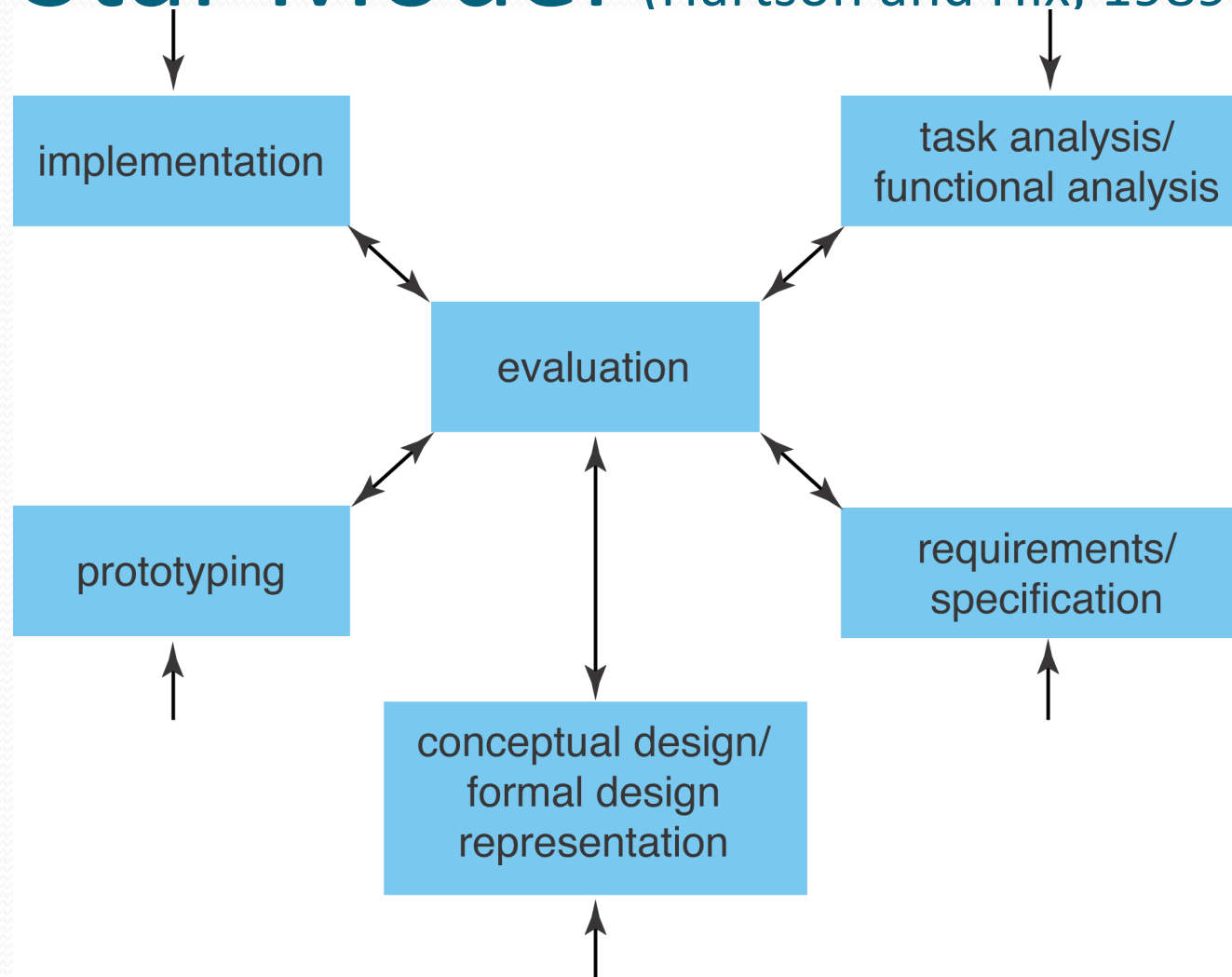
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Design as part of Usability Engineering

SE/UE lifecycle is divided into

- Definitional Activities
 - e.g. user analysis, task analysis
- Development/Design Activities
 - e.g. interaction design

The Star Model (Hartson and Hix, 1989)



The Star lifecycle model

- Suggested by Hartson and Hix (1989)
- Important features:
 - Evaluation at the center of activities
 - No particular ordering of activities. Development may start in any one
 - Derived from empirical studies of interface designers

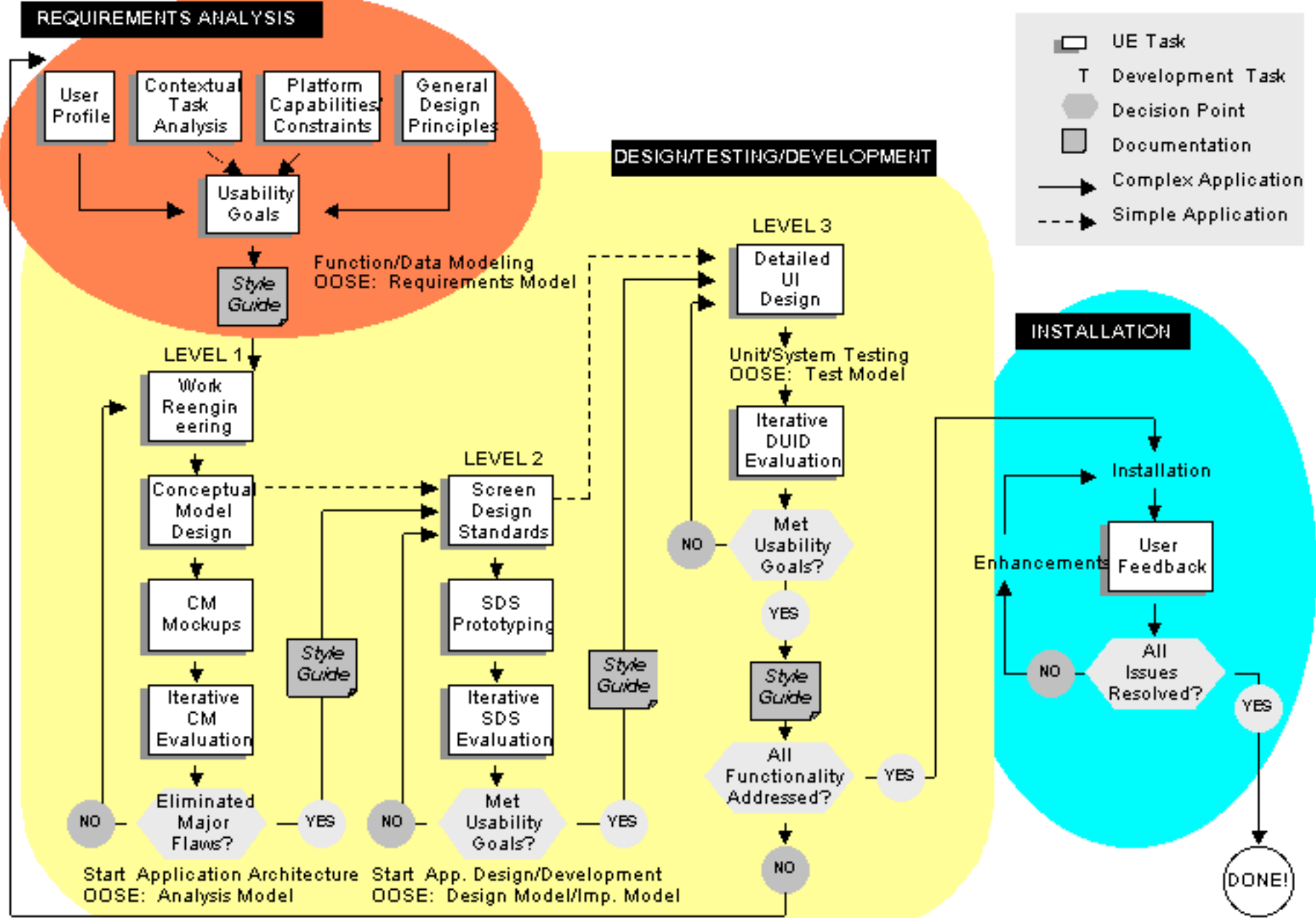
Hartson, H.R. and Hix, D. ,“Toward empirically derived methodologies and tools for human-computer interface development,
International Journal of Man-Machine Studies, v.31 n.4, p.477-494, Oct. 1989

Usability engineering lifecycle model

- Reported by Deborah Mayhew (*)
- Important features:
 - Holistic view of usability engineering
 - Provides links to software engineering approaches, e.g. OOSE
 - Stages of identifying requirements, designing, evaluating, prototyping
 - Can be scaled down for small projects
 - Uses a style guide to capture a set of usability goals

(*)D. Mayhew, *The usability engineering lifecycle: a practitioner's handbook for user interface design*, Morgan Kaufmann Publishers Inc. 1999

THE USABILITY ENGINEERING LIFECYCLE



Three types of information cointained in Style Guides

	Principles	Guidelines	Standards
Definition	General, high-level goals	General rules of thumb	Specific, customized translations
Example	Support the user's task with visual cues	Use color to code categories	Green background for status messages
Advantages	Help organize, focus design efforts	Based on research and experience, focus design	Directly applicable, tailored to users/tasks
Disadvantages	Too general, not directly applicable	Too general, require interpretation	Very specific, may not be optimal in some cases

Interaction Design (*)

- It is a process:
 - a goal-directed problem solving activity informed by intended use, target domain, materials, cost, and feasibility
 - a creative activity
 - a decision-making activity to balance trade-offs
- It is a representation:
 - a plan for development
 - a set of alternatives and successive elaborations

(*)H.Sharp,Y.Rogers,J.Preece, *Interaction Design: Beyond Human-computer Interaction*, John Wiley & Sons Ltd., (2007)

Four basic activities

There are four basic activities in Interaction Design:

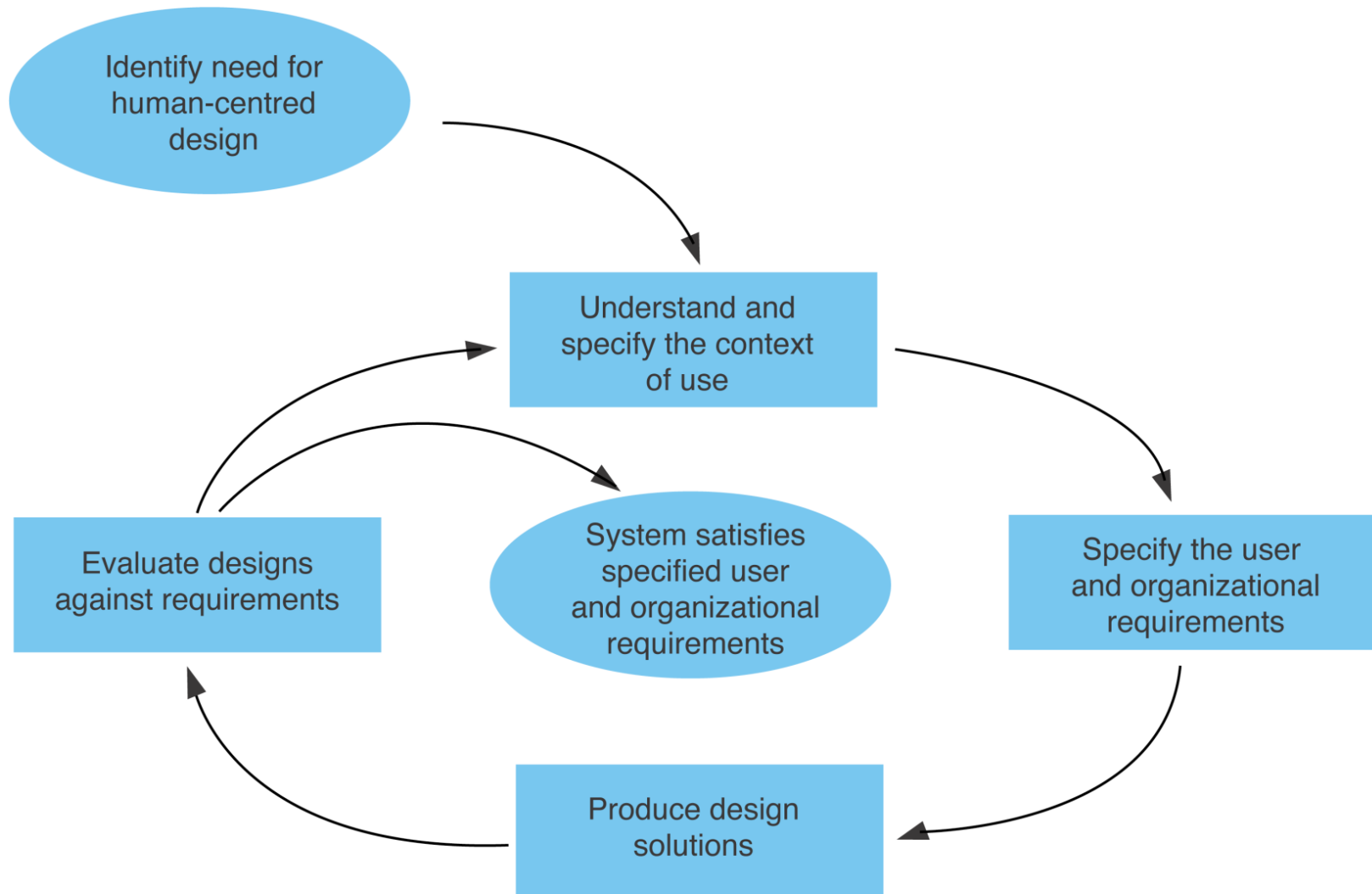
1. Identifying needs and establishing requirements
2. Developing alternative designs
3. Building interactive versions of the designs
4. Evaluating designs

Three key characteristics

Three key characteristics permeate these four activities:

1. Focus on users early in the design and evaluation of the artefact
2. Identify, document and agree specific usability and user experience goals
3. Iteration is inevitable. Designers never get it right first time

ISO 13407 - Human-centred design processes for interactive systems



Human-centred design processes for interactive systems (1999)



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User-Centred Design –from conceptual design to scenario- based design

- **Formal analysis and modeling** of well-specified tasks and of users as a means to consider human behavior and experience
- **Scenario-based design** offer lightweight methods for envisioning future use possibilities

Formal analysis and modeling - *conceptual design*

- *user modeling*
 - modeling potential users/communities of users
 - *user role modeling* especially relates design objectives to the different roles user may assume
- *task modeling*
 - *task*: 'a goal together with some procedure or ordered set of actions that will achieve that goal'
 - the task-artefact cycle (Carroll, 1990): the tasks set requirements for the development of artefacts- the use of the artefact may redefine the task that originated it
- *object modeling*
 - modeling salient interface contents
- often hybrid models have been proposed
 - *object -action model*

Task-based methods

- Different notations adopted
 - grammar type of representation
 - structure chart notation: structured hierarchical modeling of tasks and subtasks associated with users' profiles
- The MUSE method (Lim and Long,1994)
 - emphasise the role of abstraction and 'generification' in modelling
 1. Take as input task information described in the statement of requirements. Temporal and conditional aspects of task execution should also be noted.
 2. Summarise the task (and sub-tasks) in device-independent terms to reveal the logic underlying the design of the task.
 3. Re-express the task description using structured diagrams to derive a target Generalised Task Model, and record any additional notes.

Task-based methods Example

Step 1 – Input task information (from requirements)

From the system requirements:

The user approaches the kiosk to check in for a flight. They must scan their passport or enter a booking reference, select their flight, confirm passenger details, choose or change a seat, and print their boarding pass. If the passport scan fails, the user may enter details manually. The process must complete within 5 minutes.

Temporal and conditional aspects:

Task sequence: *Identify* → *Confirm* → *Choose* → *Print*

Condition: *If passport scan fails* → *manual entry task*

Temporal: *Boarding pass must be printed before the user leaves the kiosk*

Task-based methods Example

Step 2 – Summarise the task in device-independent terms

At this stage, we abstract away from *how* the task is implemented (touchscreen, keyboard, etc.) and describe the **logical structure** of the user's goals and actions.

Abstract task summary:

Goal: Obtain a valid boarding pass.

Subtasks:

Identify self (input personal/travel data)

Validate identity (system checks database)

Confirm travel details (user reviews and accepts)

Select options (seat, baggage, etc.)

Produce confirmation (boarding pass issued)

Each subtask could later be performed on various devices (kiosk, mobile app, web form), maintaining the same logical model.

Task-based methods Example

Step 3 – Re-express using structured diagrams → Generalised Task Model

You can use **Hierarchical Task Analysis (HTA)** or **ConcurTaskTrees (CTT)** notation to model the structure.

Example (simplified **hierarchical diagram** form):

o. Check-in for flight

1. Identify self

1.1 Scan passport [OR]

1.2 Enter booking reference

2. Validate identity

3. Confirm travel details

4. Select options

4.1 Choose seat

4.2 Add baggage (optional)

5. Produce boarding pass

5.1 Display summary

5.2 Print boarding pass

Notes:

Temporal constraint: Steps 1–5 must be sequential.

Conditional path: Step 1.1 *fails* → perform 1.2.

Parallel possibility: Step 4.1 and 4.2 can occur in any order.

This **Generalised Task Model (GTM)** captures the logic of the task in an abstract, reusable way — independent of any particular device or interface design.

Object-based model

- Synthesizing objects and actions
 - the *what* and *how* of users' tasks
 - objects come directly from users' tasks
 - e.g. articles, words/phrases, passages, sections
 - e.g. tracks, albums, length
 - Properties of objects
 - distinguish instances of objects
 - Actions are methods the users use to do the tasks
 - e.g. open/read, close/write, modify, use, delete



Object-based model -exercise

- Consider again the **airport check-in kiosk example**, and use the **Object Model** approach, which focuses on **what** the users work with (objects) and **how** they interact with them (actions or methods).

Design as a process supporting usability

Design is inherently creative and unpredictable. Interactive system designers must blend a thorough knowledge of technical feasibility with a mystical esthetic sense of what attracts users (Schneiderman and Plaisant, 2005).

Rosson&Carroll (2002) characterize design as:

- Design is a *process*; it is not a state
- The design process is *nonhierarchical*;
- The process is radically *transformational*;
- The design intrinsically involves the *discovery of new goals*

Discovery of new goals

- The dynamic nature of design may lead to
 - an iterative refinement/modifications of user's requirements and to
 - the consequent *discovery of new goals*
- the lighter is the representation method adopted to envision present and future use possibilities
- the more efficient and effective is the achievement of usability results

A scenario-based design process

(Rosson & Carroll 2002)

